Scale-Aware Detailed Matching for Few-Shot Aerial Image Semantic Segmentation

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***Abstract*— Few-shot semantic segmentation, aiming to segment query images with a few annotated support samples, has drawn increasing attention. Most existing few-shot methods leverage the single prototype obtained from global average pooling to represent all support information and further use the extracted prototype to segment the query images in a matching manner. Although promising results for natural images have been reported, these methods cannot be directly applied on aerial images. The main reason comes from that the extracted single support prototype can only provide a coarse guidance for matching between query and support images and could not handle the large variance of objects' appearances and scales. To deal with these challenges on aerial images, we propose a scale-aware few-shot semantic segmentation network to perform detailed matching with multiple prototypes. More specifically, the detailed matching module is first constructed to compute the pixel-level similarity between the query features and the extracted multiple support prototypes for providing more accurate parsing guidance. Subsequently, to address the problem of scale imbalance, the scale-aware focal loss is designed to dynamically down-weight the loss assigned to large well-parsed objects and focus training on tiny hard-parsed objects. To facilitate the reproducible research on the task of few-shot semantic segmentation in aerial images, we further provide a few-shot segmentation benchmark iSAID-5^i constructed from the large-scale iSAID dataset [1]. Comprehensive experiments and comparisons with the state-of-the-art few-shot segmentation methods on the iSAID-5^i dataset clearly demonstrate the superiority of our proposed method. The code and dataset are available at https://github.com/caoql98/SDM.**

***Index Terms*— Detailed matching match, few-shot semantic segmentation, scale-aware focal loss.**

I. Introduction

With the increasingly rapid development of modern remote sensor technologies, semantic segmentation in remote sensing images has become an essential task for various applications, such as land-use and land-cover mapping [2]–[6], forest management [7], [8], and building/road detection [9], [10]. Recently, deep learning technology has achieved encouraging performance for the task of aerial image semantic segmentation [11]–[23], and most deep-learning-based segmentation models heavily depend on large-scale annotated datasets, such as DLRSD [24] and WHDLD [25]. However, acquiring such large quantities of pixel-level labeled datasets is laborious and expensive. Meanwhile, it is irrational for us to get abundant annotated data of all diverse semantic objects. To tackle this dilemma, the few-shot semantic segmentation task, which is defined to precisely parse the query images with only a few annotated support samples, has shown great potential values for various applications.

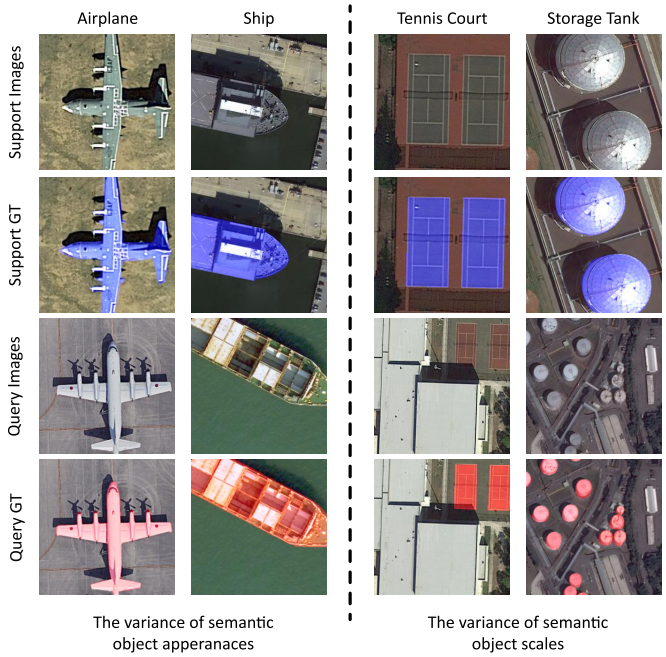


Fig. 1. Variances of semantic objects in aerial images.

The current few-shot semantic segmentation algorithms [26]–[32] in nature images usually regard the segmentation as a foreground detection task by performing the global comparison between the query features and the support prototypes. For instance, the SG-one network [33] first proposes the support prototype to guide the parsing process of query images. Following this research line, some works [26], [30], [34] attempt to design more representative support prototypes or to capture more complete query information for boosting the segmentation performance.

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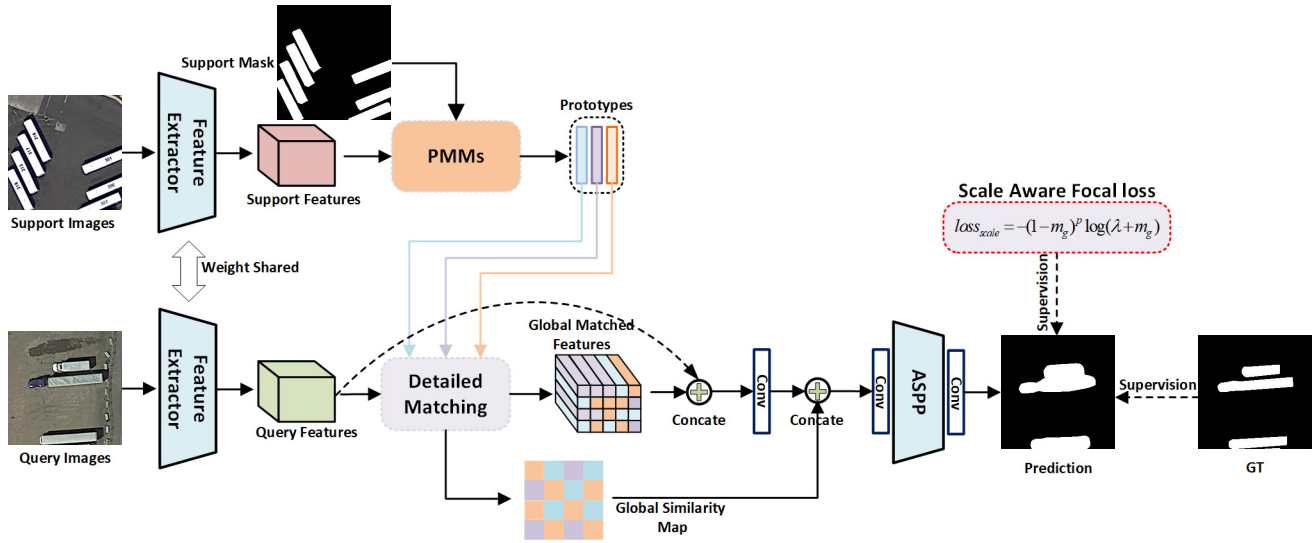


Fig. 2. Overall pipeline of our proposed network. The pretrained VGG16 [39] or Resnet50 [40] is leveraged as the feature extractor to generate the corresponding query and support features. Then, we adopt the PMMs [32] to obtain multiple prototypes from support features. Based on the multiple prototypes, the detailed matching module would generate the global matched features and the global similarity map to provide accurate pixel-level parsing guidance. We further design a scale-aware focal loss to address the scale imbalance with the scale adaptability for boosting the parsing performance.

Although remarkable progresses have been achieved for natural images, these methods fail to effectively tackle the few-shot segmentation task on aerial images. The main reason is that the single support prototype obtained from global average pooling only provides a coarse guidance for foreground detection in query images and could not handle the large variance of the semantic objects' appearances and scales in aerial images. More specifically, as shown in the left of Fig. 1, there is a huge variance of objects' appearances, e.g., the shape of airplanes and the structures of ships, between the query and support images. Thus, using the single prototype regardless of the pixel-level appearance characteristics would lead to obvious semantic ambiguity for the matching between the query and support images. Meanwhile, just like the images in the right of Fig. 1, the tennis court and storage tank categories have large differences in terms of scale. This obvious scale imbalance could not be addressed using the single support prototype. Accordingly, this unresolved dilemma would inevitably lead to the domination of the large easy-parsed objects in the predicted results, and the tiny hard-parsed objects would likely be ignored.

To overcome the aforementioned limitations of few-shot segmentation algorithms in aerial images, we propose a scale-aware few-shot semantic segmentation network to perform detailed matching with multiple prototypes. Particularly, a novel detailed matching module is first designed to precisely match the query features and the diverse support prototypes at pixel level. First, multiple prototypes are generated based on the support features and are further compared with the query features to compute the pixel-level similarity maps. Using these similarity maps as the selection guidance, the most similar prototype at each pixel location of query feature maps is aggregated to generate the global matched features. Simultaneously, the pixel-level similarity maps are further summed to obtain the global similarity map. By jointly concatenating the query features with the global matched features and the global similarity map, we maximally provide accurate parsing guidance.

Although the detailed matching module provides accurate parsing guidance, the tiny object could still be easily neglected when the support and query have large difference in terms of scale. Thus, inspired by the focal loss [35], the scale-aware focal loss is further proposed to constrain the domination of easy-parsed large objects and improve the focus on tiny objects during the training process. To directly measure the scale of predicted semantic objects, we sum the probability of each pixel in the predicted mask to get the scale score. Naturally, the small scale score would indicate two situations that the scale of semantic objects is truly small or the predictions fairly stay away from the ground truth. Both the situations mean that this sample is a hard-parsed sample. Based on this fact, the scale-aware focal loss would become tremendous with minute predicted object sizes and decay to near 0 as confidence in the large objects. Intuitively, the designed scale-aware focal loss could automatically down-weight the contribution of huge easy-parsed objects and rapidly arouse the attention of the hard and inconspicuous samples during the training phase.

The current semantic segmentation algorithms mainly perform experiments on Potsdam and Vaihingen datasets, which are both provided by the ISPRS II/4 committee [36] and some works would adopt other specific-task datasets such as DLRSD [24] and WHDLD [25]. However, few-shot semantic segmentation task requires abundant base classes' images and diverse semantic object categories for base/novel classes' split. Thus, the current available semantic segmentation datasets for remote sensing images could not satisfy the base requirement because of their limited scale or classes of datasets. Compared with the previous datasets, the newly opened iSAID dataset [1] provides quantities of remote sensing images with 15 different categories of semantic objects, which exactly match the demand of few-shot segmentation. Based on this dataset, we further provide the iSAID-5^i dataset which is

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elaborately designed for few-shot semantic segmentation task in the remote sensing domain.

The cooperation of the detailed matching module and the scale-aware focal loss formulates a novel few-shot segmentation network of which the detailed matching provides accurate parsing guidance and the scale-aware focal loss arouses the focus on tiny hard-parsed objects during the training process. Extensive experiments on the iSAID-5^i dataset clearly demonstrate the superiority of our network. The main contributions of our method are summarized as follows.

1) The detailed matching module is designed to provide accurate pixel-level matching between the query and support. Benefiting from this module, our network maximally provides the accurate guidance for the segmentation process.
2) The scale-aware focal loss is proposed to down-weight the loss assigned to huge well-parsed objects and arouse the focus on tiny hard-parsed objects, where the hard-parsed example mining is naturally achieved.
3) A large-scale iSAID-5^i dataset is elaborately built for few-shot semantic segmentation task. The experimental results on the dataset demonstrate the splendid effectiveness of our designed model.

II. Related Works

In this section, we will review the related works about semantic segmentation on remote sensing domain and introduce the popular few-shot segmentation algorithms on nature images.

A. Aerial Image Semantic Segmentation

Semantic segmentation is a fundamental yet challenging task aiming at providing a dense prediction of class at pixel level. Recently, with the development of convolutional neural networks (CNNs), aerial image semantic segmentation has obtained an increasing interest [11], [12], [14], [16]–[21]. Some works focus on using the combination of CNNs and the statistical models to address this task. For instance, the work [37] uses conditional random fields to assist the deep CNNs with more precise per-pixel class probabilities, and Zhang *et al.* [38] incorporate the Markov random field to provide crisp and accurate boundary delineation. Some other works try to solve this task using multi-scale feature fusion structure. Among them, Yu *et al.* [20] first introduce the pyramid pooling module to tackle the semantic segmentation on aerial images. Meanwhile, Ding *et al.* [12] propose a two-stage scheme to leverage the features from different layers to boost the segmentation performance Similarly, Liu *et al.* [14] implement multiple weighted edge supervisions on the outputs to dig out useful semantic information. Furthermore, Marmanis *et al.* [15] propose a segmentation network to improve the parsing performance with boundary detection. With the rising of multiple attention mechanisms, Ding *et al.* [11] design the efficient local attention embedding to reinforce the segmentation performance. Simultaneously, Panboonyuen *et al.* [17] combine the global convolutional network with the channel attention to gather more discriminative features. The exploitation of additional training information has also been introduced to address the semantic segmentation of remote sensing images, such as the utilization of open street maps in [13] and [41] and the introduction of the digital surface model (DSM) in [19].

B. Few-Shot Semantic Segmentation

The few-shot semantic segmentation task aims at parsing novel semantic objects on query images with a few annotated support images. By first introducing the prototype concept for few-shot segmentation, Zhang *et al.* [33] leverage the similarity comparison between the prototypes and query images to achieve the few-shot segmentation task. Simultaneously, some works mainly focus on designing better prototypes to solve this problem. Among them, Wang *et al.* [30] propose the prototype alignment regularization to obtain more generalized prototypes which significantly improve the parsing performance. Furthermore, PMMs [32] design a parameter-free method to generate multiple prototypes and leverage diverse prototypes to boost the segmentation performance with more complete feature representations. However, the hierarchical information is neglected in this network. To address this issue, a multilevel comparison model is proposed in CAnet [27] to optimize the parsing result iteratively. The knowledge transfer challenge from base categories to new categories is not addressed in the aforementioned models. To conquer that, the new and old knowledge are combined in the AMP [26] to iteratively update the adaptive masked proxies' features to perform effective segmentation. Meanwhile, PFEnet [31] attempts to solve the transfer challenge between novel and base classes by the prior knowledge. Instead, PGnet [30] builds a pyramid graph network to perform precise few-shot segmentation. Although these methods achieve remarkable progresses on natural images, all previous methods lack the adaptability on aerial images since the extracted single support prototype can only provide a coarse guidance for matching between the query and support images and could not handle the large variance of objects' appearances and scales.

III. Proposed Method

A. Problem Setting

The goal of the few-shot semantic segmentation system is to discover the foreground pixels in query images with a few annotated samples from support set. Suppose we split the abundant data into two non-overlapping classes set C_{base} and C_{novel}, then the training dataset D_{train} is built on the class set C_{base} and the test dataset D_{test} is built on the class set C_{novel}.

In the $k-$shot setting, we episodically sample $k+1$ annotated images $\{(I_s^1, M_s^1), (I_s^2, M_s^2), \ldots, (I_s^k, M_s^k), (I_q, M_q)\}$ in the targeted class from the training dataset D_{train}, where (I_s^i, M_s^i) denotes the support samples, and (I_q, M_q) means the query pairs. During the training stage, the support samples (I_s^i, M_s^i) and the query image I_q are propagated into the segmentation model to acquire the prediction mask $\widetilde{M}_q$, which is supervised by the ground truth M_q. Similarly, during the test stage, we, respectively, sample $k+1$ annotated images in different novel classes from the dataset D_{test} and leverage

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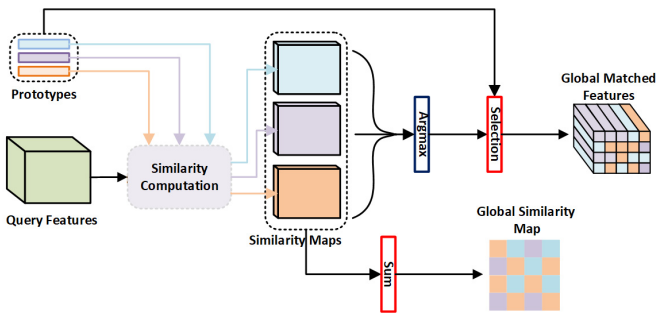


Fig. 3. Illustration of the detailed matching module.

the labeled $k-$shot support images to inference the semantic objects in the query images.

B. Method Overview

As shown in Fig. 2, we propose a novel and effective few-shot semantic segmentation network by providing accurate pixel-level matching between the query and support and arouse the attention in tiny hard-parsed objects during the training stage. More specifically, we first use the pretrained backbones (VGG16 [39] or Resnet50 [40]) as the feature extractor to generate the corresponding query and support features. Then, the PMMs [32] are leveraged on the support features to obtain multiple prototypes. Different from [32] that only gives a coarse parsing guidance, as illustrated in Fig. 3, we design the detailed matching module to perform the pixel-level matching between the query features and support prototypes for providing more accurate parsing guidance. With the global matched features and the global similar map generating from the detailed matching module, we, respectively, fuse them with the query features. Subsequently, the fused features are propagated into the designed convolutional layers with the Atrous Spatial Pyramid Pooling (ASPP [42]) to predict the final parsing result. Although the detailed matching module provides accurate parsing guidance, the tiny object could still be easily neglected when the scales of the support and query objects are largely different. Thus, to tackle the scale imbalance problem, the scale-aware focal loss is designed to constrain the domination of large easy-parsed objects and arouse the focus on tiny hard-parsed objects by dynamically performing hard-parsed samples' mining.

C. Detailed Matching Module

Most prior algorithms only use single prototype to provide coarse parsing guidance with an ambiguous match between the query and support. To break this dilemma, we design the detailed matching module to precisely match the query features and the diverse support prototypes at pixel level. The whole architecture of the detailed matching module is illustrated in Fig. 3.

Assume that we have query features F_q and support features F_s, where the size of F_q and F_s equals $C \times H \times W$, we would first adopt the PMMs [32] on the support features F_s to generate multiple prototypes $\{P_1, P_2, \ldots, P_N\}$, where $P_1 \in \mathbb{R}^{C\times 1}$. Specifically, the Gaussian mixture models are first constructed on the support features and are further iteratively updated by performing the expectation–maximization (EM) algorithm. The distribution vectors of the final generated Gaussian mixture models are regarded as multiple prototypes $\{P_1, P_2, \ldots, P_N\}$. Then, the cosine similarity $s_i^{x,y}$ between each prototype and each element of query features is computed as

$$s_i^{x,y} = \frac{P_i \cdot F_q^{x,y}}{\|P_i\| \cdot \left\|F_q^{x,y}\right\|} \tag{1}$$

where $F_q^{x,y} \in \mathbb{R}^{C\times 1}$ is the corresponding feature vector at the location (x, y) of the query features, and $\|\cdot\|$ denotes the Euclidean norm.

By collecting the cosine similarity $s_i^{x,y}$ at all locations, we would obtain the pixel-level similar map $M_i \in \mathbb{R}^{H\times W}$ for each prototype P_i. Subsequently, to perform the accurate match at pixel level, the argmax operation is applied on N similar maps to acquire the index $I^{x,y}$ of the most similar prototype for each element of the query features

$$I^{x,y} = \arg\max_{i\in[1,N]} M_i^{x,y} \tag{2}$$

where $M_i^{x,y}$ means the value at the location (x, y) of the similar maps, and N is the number of the similarity maps.

By leveraging the index $I^{x,y}$ as the pixel-level guidance, we would precisely construct the global match features $F_m \in \mathbb{R}^{C\times H\times W}$ by inserting the most similar prototype at the corresponding location. In this way, the global match features would provide the most precise guidance for the subsequent parsing process. Simultaneously, to further boost the guidance with a comprehensive understanding of the global similarity between the query and support, the similarity maps M_i would be summed to obtain the global similarity map $M_c \in \mathbb{R}^{1\times H\times W}$ to perform accurate parsing

$$M_c = \sum_{i=1}^{N} M_i \tag{3}$$

where N is the number of the similarity maps.

To effectively use accurate pixel-level guidance generated from the detailed matching module, we would, respectively, fuse the query features F_q with the global matched features F_m and the global similarity map M_c. Particularly, we would directly concatenate the query features F_q and the global match features F_m in channel dimension and further leverage a convolutional layer to reduce the number of channels

$$F_q^h = \text{conv}(\text{concate}(F_m, F_q)) \tag{4}$$

where F_q^h denotes the matched query features. Furthermore, the global similarity map M_c would be concatenated with F_q^h to provide a more accurate global guidance. Then, the concatenated features would be propagated into the designed convolutional layers with the ASPP [42] to predict the final parsing result R_f

$$R_f = \text{ASPP}\left(\text{concate}\left(F_q^h, M_c\right)\right). \tag{5}$$

In the ASPP module, the concatenated features would be first propagated into four atrous convolutional layers with

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TABLE I
DISTORTION CLASSES IN iSAID-5^i DATASET

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
ship	storage tank	baseball diamond	tennis court	basketball court	ground track field	bridge	large vehicle	small vehicle	helicopter	swimming pool	roundabout	soccer ball field	plane	harbor

different dilation rates to generate four corresponding features. Then, by combining these features with the concatenation and a convolutional layer, we could get the final parsing result R_f. The final parsing result R_f is supervised by the ground truth with the cross-entropy loss.

D. Scale-Aware Focal Loss

Although the detailed matching module provides accurate parsing guidance, the tiny hard-parsed object could still be easily neglected when the support and query have large-scale difference. This phenomenon would inevitably result in a weak segmentation model that lacks the ability to detect and parse the tiny semantic objects. When processing the aerial images, this issue would get worse due to the fairly huge scale gap of the semantic objects. Unfortunately, most existing few-shot segmentation algorithms ignore such a vita problem.

Thus, to conquer the scale imbalance, a straightforward method is to leverage the scale of the semantic objects in ground-truth masks as a guiding score to eliminate the domination of salient and large semantic objects. However, such a static adjusting strategy would lead the model to another suboptimal corner since the focus on tiny objects might be overly aroused and the scale-aware ability would be fixed on a single pattern.

Instead of using the scale of the semantic objects in the predicted mask as the guiding score, a scale-aware focal loss is designed to dynamically relieve the domination of large well-parsed objects and arouse the focus on tiny hard-parsed objects. The tiny semantic objects in the predicted mask would indicate two situations that the scale of semantic objects is truly small or the predictions are fairly staying away from the ground truth. Both the situations mean that this training sample is a hard-parsed sample. The focus on this query–support pair should be aroused to help the model get a more generalized parsing ability. From the aforementioned analysis, we could figure out that that the scale of semantic objects in predict masks could flexibly reflect the adaptability of diverse semantic objects and help the segmentation model become more generalized.

Based on this assumption, let the predicted mask be $m_p \in \mathbb{R}^{H\times W}$, we could compute the scale score m_g by directly reweighting the sum of m_p with the images' scale since each location of m_p denotes the pixel-level foreground confidence, and a small scale score illustrates tiny semantic objects

$$m_g = \frac{1}{HW}\sum_{i,j} m_{p,i,j} \tag{6}$$

with the scale score m_g generating, and the scale-aware focal loss is designed as follows:

$$\text{loss}_{\text{scale}} = -(1-m_g)^p \log(\lambda + m_g) \tag{7}$$

TABLE II
TESTING CLASSES FOR THREEFOLD CROSS VALIDATION TEST. THE TRAINING CLASSES OF iSAID-5^i, $i = 0, 1, 2$ ARE DISJOINT WITH THE TESTING CLASSES

Dataset	Test classes
iSAID-5^0	ship, storage tank, baseball diamond, tennis court, basketball court
iSAID-5^1	ground track field, bridge, large vehicle, small vehicle, helicopter
iSAID-5^2	swimming pool, roundabout, soccer ball field, plane, harbor

where p and λ are the hyperparameters, of which λ is a very small number, and the study of the parameters is illustrated in Section IV.

Note that when m_g approaches 0, the scale-aware focal loss would increase rapidly which could dynamically force the network to focus on the hard-parsed samples.

Finally, given the supervision of segmentation ground truth as cross-entropy loss loss_{GT}, the overall supervision loss is defined as

$$\text{loss}_{\text{all}} = \text{loss}_{\text{GT}} + \mu^* \text{loss}_{\text{scale}} \tag{8}$$

where μ is the adjusting factor, and the factor is empirically set as 0.5 in this article.

IV. EXPERIMENTS

To demonstrate the effectiveness of the proposed method, the experiments are organized as follows. We first describe the newly created dataset iSAID-5^i. Next, the implementation details and the parameter analysis are provided. Then, the segmentation results and comparisons with the state-of-the-art few-shot segmentation methods are presented. We finally present a series of ablation studies to demonstrate the impact of each component in our proposed few-shot segmentation algorithm.

A. Datasets

The existing available aerial image semantic segmentation datasets for the few-shot segmentation task mainly have two drawbacks: the limited numbers and classes of semantic objects, which indicate that these datasets could not satisfy the base requirement of the few-shot segmentation task. Instead, the newly opened iSAID dataset [1] provides a feasible solution for us. It contains 655 451 object instances for 15 categories across 2806 high-resolution images, which exactly match the requirement of the few-shot segmentation task. Based on this, we follow the setting in PASCAL-5^i [33] to create the iSAID-5^i dataset, and the class details are shown in Table I. Particularly, for the 15 object categories in the iSAID-5^i dataset, the cross-validation method is leveraged

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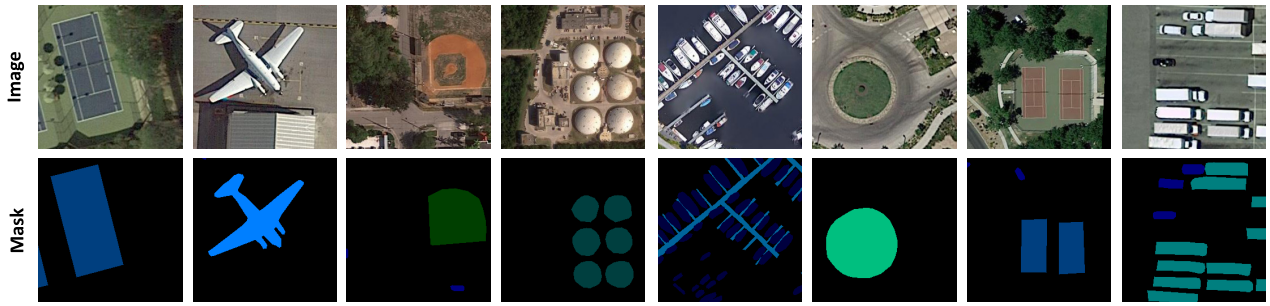


Fig. 4. Illustration of samples in iSAID-5^i.

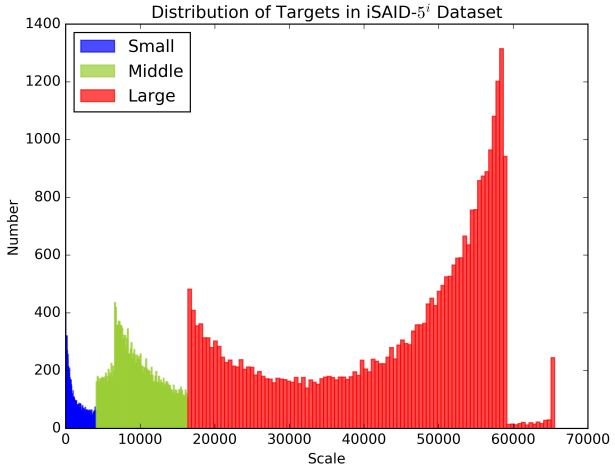


Fig. 5. Distribution of targets in the iSAID-5i dataset.

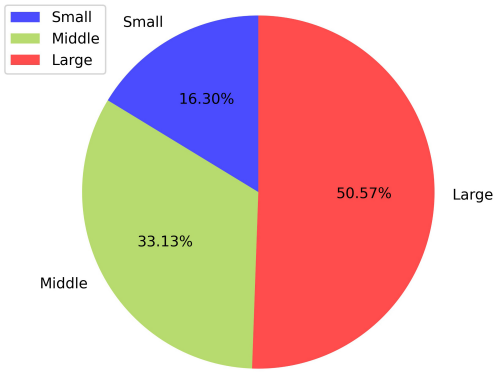


Fig. 6. Proportion of targets in the iSAID-5i dataset.

to evaluate the proposed model by sampling five classes as test categories D_{test} and leveraging the left ten classes as the categories of the training set D_{train}. The details of the class splits are shown in Table II, where I is the fold number. To further enlarge the available labeled data, we randomly crop images in 256×256 from the original images to rebuild the iSAID-5^i dataset. This dataset contains 18 076 images for training and 6363 images for validation, and some examples of iSAID-5^i are illustrated in Fig. 4. In the $k-$ shot setting, $k+1$ images would be randomly sampled from the training set D_{train}, and each image has the same category label. Among the $k+1$ images, k images are fed into the parsing model as support images accompanied by its annotated masks. Another image is regarded as the query image and its annotated mask is treated as the ground truth to calculate the supervision loss. Aiming at analyzing the scale distribution of the instances in the constructed iSAID-5i dataset, we collect the scales of diverse instances with the GT mask for instance segmentation. The corresponding distribution and proportion of the targets are, respectively, shown in Figs. 5 and 6, where the range of the small, middle, and large targets' size is, respectively, defined as $(0, 64 \times 64]$, $(64 \times 64, 128 \times 128]$, and beyond 128×128. Focusing on the results in Fig. 5, we could clearly figure out that the size of the annotated targets varies rapidly and the constructed dataset provides abundant annotated samples at each size of targets. Furthermore, as can be seen from Fig. 6, the large targets occupy the largest proportion of the constructed dataset, the middle targets follow, and the small targets make up the smallest proportion of the constructed dataset. If we scratch a little deeper about the size of targets, we could see that small, middle, and large targets all have sufficient size ranges and abundant annotated samples.

Thus, the constructed iSAID-5i dataset provides sufficient size diversity for the few-shot aerial images' semantic segmentation task.

B. Implement Details

In our work, the class mean intersection over union (mIOU) is used as the evaluation metric. Our network leverages the VGG16 [39] and Resnet50 [40] networks pretrained in Imagenet as backbones. The overall network is built on Pytorch. The proposed parsing model is trained end to end with SGD optimizer where the learning rate is $3.5e-4$, the weight decay is $5e-4$, and the momentum is 0.9. The training process lasts for 50 000 iterations until convergence. The training images are propagated into the segmentation model with the size of 256×256. All the training images are augmented with random horizontal flipping. The parameters p and λ are, respectively, set as 1 and 0.1 during the training procedure, which are further studied in the ablation study. To better testify the superiority of our proposed segmentation model, we, respectively, perform a series of performance comparison experiments in three splits of iSAID-5^i at one-shot and five-shot settings.

C. Performance Analysis

The comparisons between our model and other popular methods with the available source codes in the iSAID-5^i

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TABLE III

PERFORMANCE COMPARISONS OF DIFFERENT METHODS WITH DIFFERENT SPLITS ON THE iSAID-5^i DATASET

Method	1-shot				5-shot			
	Split-0	Split-1	Split-2	Mean	Split-0	Split-1	Split-2	Mean
VGG-16								
PAnet [30]	17.43	11.43	15.95	14.94	17.70	14.58	20.70	17.66
CAnet [27]	19.73	17.98	30.93	22.88	23.45	20.53	30.12	24.70
PMMs [32]	20.87	16.07	24.65	20.53	23.31	16.61	27.43	22.45
PFEnet [31]	16.68	15.30	27.87	19.95	18.46	18.39	28.81	21.89
Ours	**29.24**	**20.80**	**34.73**	**28.26**	**36.33**	**27.98**	**42.39**	**35.57**
Resnet50								
PAnet [30]	12.36	9.11	12.05	11.17	13.82	12.40	19.12	15.11
CAnet [27]	18.80	15.62	25.79	20.07	23.86	18.54	32.00	24.80
PMMs [32]	19.02	18.51	28.42	21.98	20.89	20.87	31.23	24.33
PFEnet [31]	18.75	17.24	22.09	19.36	19.57	18.43	26.14	21.38
Ours	**34.29**	**22.25**	**35.62**	**30.72**	**39.88**	**30.59**	**45.70**	**38.72**

TABLE IV

PERFORMANCE COMPARISONS OF DIVERSE CLASSES ON THE iSAID-5^i DATASET WITH ONE-SHOT SETTING

Methods	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
VGG-16															
PAnet [30]	13.85	17.78	19.78	17.80	17.94	**22.41**	11.75	11.75	6.36	4.88	12.96	19.11	23.49	11.87	12.33
CAnet [27]	7.51	12.87	31.22	30.45	16.61	18.31	**31.94**	23.15	3.77	12.55	**33.99**	**51.41**	36.69	17.61	14.91
PMMs [32]	12.84	23.06	27.81	26.65	13.97	18.48	13.76	25.58	**11.68**	10.88	28.52	34.62	34.15	16.32	9.64
PFEnet [31]	3.94	4.70	30.95	29.21	14.59	15.01	14.63	27.68	10.13	9.06	29.40	47.89	25.23	25.20	11.63
Ours	**26.55**	**30.57**	**33.01**	**35.06**	**21.03**	20.54	28.89	**32.45**	9.62	**12.59**	30.79	43.52	**49.47**	**27.51**	**22.37**
Resnet50															
PAnet [30]	7.27	9.99	14.49	13.20	16.85	**17.40**	10.23	7.37	6.21	4.33	8.35	12.38	21.28	8.52	9.75
CAnet [27]	7.25	27.25	**43.32**	6.17	9.99	6.16	17.81	28.58	8.84	16.73	17.73	56.97	13.82	27.41	13.06
PMMs [32]	12.93	17.92	41.97	7.43	14.87	13.24	32.67	26.08	4.17	16.37	12.26	**61.49**	38.37	17.92	12.02
PFEnet [31]	2.98	6.30	**33.29**	34.49	16.71	8.68	8.67	34.62	**18.19**	16.08	**28.58**	**24.47**	7.37	39.76	10.31
Ours	**37.66**	**34.37**	34.45	**39.81**	**25.14**	16.77	**34.53**	**30.50**	12.42	**17.02**	20.69	56.83	**42.80**	**40.52**	**17.26**

dataset are shown in Table III, and we highlight the best performance in each comparison. It is clear that our model outperforms the current few-shot segmentation models and acquires better performance improvement with the reinforcement of backbones. Particularly, when comparing the mean mIOU performance with the VGG16 backbone, the proposed method acquires 5.38% mIOU improvement in the one-shot setting and 10.87% mIOU improvement in the five-shot setting. Simultaneously, with Resnet50 backbone, our model achieves 8.74% mIOU improvement in the one-shot setting and 13.92% mIOU improvement in the five-shot setting. Furthermore, when using VGG16 as backbone in the one-shot scenario, the best performance boost is obtained in split 0, where our model could acquire 8.37% performance improvement. Meanwhile, we could figure out the same result with the five-shot scenario, i.e., our proposed model gets 12.88% mIOU improvement at split 0, which is exactly the best performance boost across diverse splits. When focusing on the results with Resnet50 backbone, there is a correspondingly similar result that the best performance boost is obtained in split 0. This homologous phenomenon indicates that split 0 is relatively easier for transfer learning from base classes to novel classes. From the above comprehensive analysis, we could draw a conclusion that our proposed method could effectively tackle the few-shot semantic task on aerial images.

To further analyze the performance on diverse classes with few-shot setting, we collect the detailed results with the one-shot and five-shot scenarios, and the results are, respectively, illustrated in Tables IV and V. In Table IV, it is an obvious result that our proposed method gets excellent results in almost all the classes. Particularly, the pleasant performance, namely, 26.55% mIOU and 13.71% mIOU improvement, is achieved in class of ship (C1) with VGG16 backbone. It is obvious that this superiority gets aroused with the enhancement of backbones. Specifically, with Resnet50 backbone, our model could get 37.66 mIOU and 24.73% mIOU improvement in ship class (C1).

In Table V, the increase in the number of support samples further improves the superiority of our proposed model. Specifically, with VGG16 backbone, our model gets 20.88% mIOU improvement in plane category (C14), which is the best performance boost across all the classes. With Resnet50

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TABLE V

PERFORMANCE COMPARISONS OF DIVERSE CLASSES ON THE iSAID-5^i DATASET WITH FIVE-SHOT SETTING

Methods	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
VGG-16															
PAnet [30]	13.40	17.79	19.04	18.11	20.15	22.97	16.18	15.65	9.37	8.73	31.65	21.63	25.36	12.70	12.15
CAnet [27]	11.87	18.36	33.26	26.05	27.69	19.20	29.93	30.37	11.76	11.37	23.76	60.25	28.67	21.77	16.16
PMMs [32]	14.99	25.09	28.49	30.40	17.58	18.37	13.40	27.12	12.45	11.68	35.46	36.57	37.08	18.41	9.63
PFEnet [31]	2.64	8.57	29.09	30.76	21.26	15.22	21.58	31.16	12.06	11.90	33.56	48.47	24.92	25.29	11.79
Ours	**30.81**	**38.59**	**37.34**	**42.58**	**32.35**	**32.32**	**39.71**	**37.23**	**17.73**	**12.93**	**39.20**	**66.78**	**39.06**	**46.17**	**20.72**
Resnet50															
Panet [30]	8.45	11.68	17.81	14.22	16.93	18.86	15.70	9.40	7.90	10.13	21.41	24.94	28.41	9.80	11.04
Canet [27]	32.47	6.49	39.8	25.97	14.59	25.01	25.27	32.06	1.31	9.08	26.99	73.84	33.83	20.69	4.69
PMMs [32]	13.44	22.22	42.12	8.58	18.06	13.17	37.88	30.69	6.41	**16.21**	14.62	65.02	42.99	20.86	12.66
PFEnet [31]	10.13	9.48	30.71	31.23	16.31	11.53	14.15	36.07	15.58	14.82	**38.52**	20.90	20.41	38.66	12.22
Ours	**38.76**	**49.06**	**50.06**	**39.25**	**22.26**	**30.68**	**45.34**	**41.49**	**20.21**	15.21	32.61	**66.64**	**57.41**	**49.12**	**22.71**

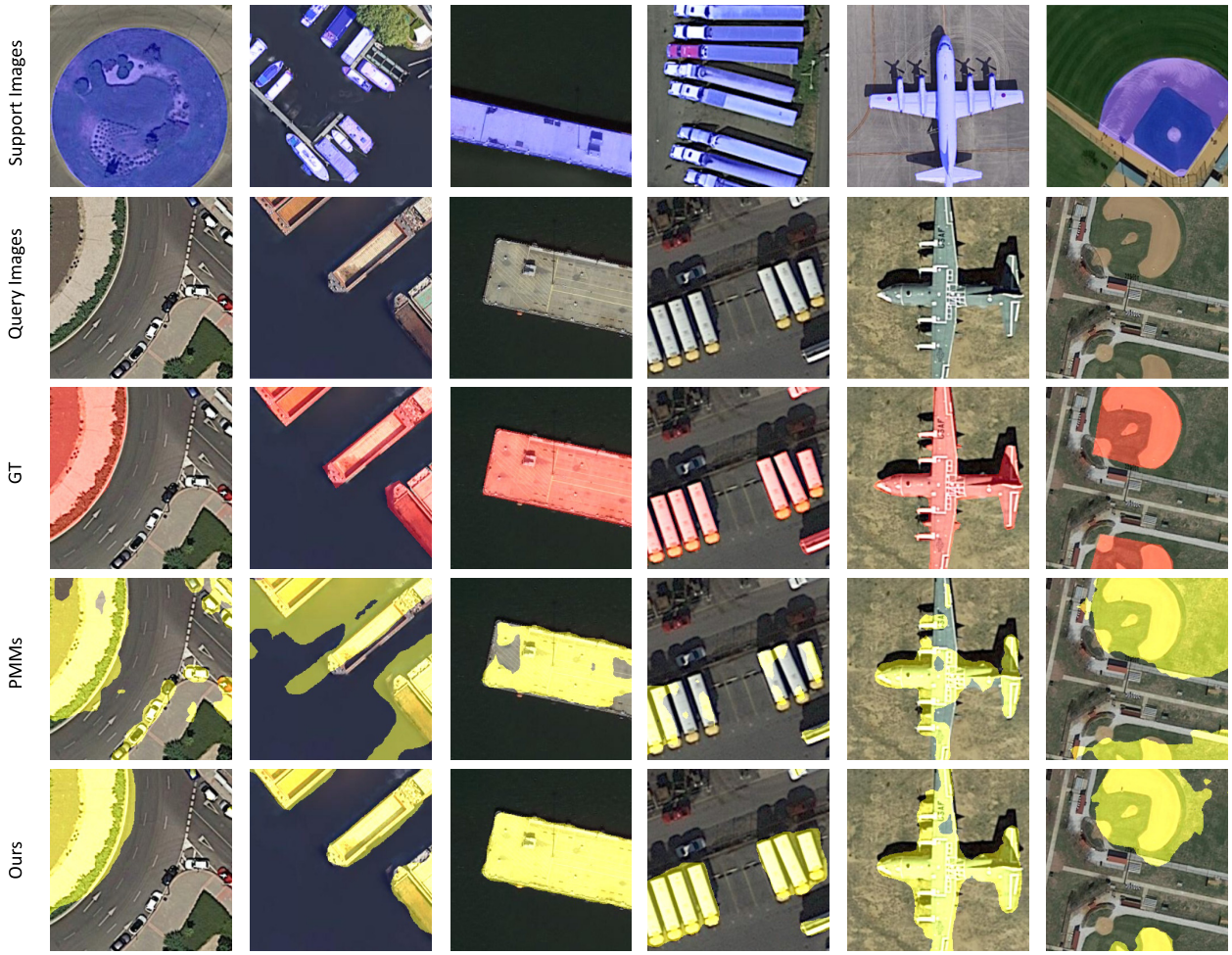


Fig. 7. Qualitative results of our model. (Top–bottom) Support images, query images, ground truth of query images, predictions of PMM network, and predictions of our network. The last two columns denote false results.

backbone, our model gets the best performance boost, namely, 26.84% improvement, in storage tank class (C2). Furthermore, the best parsing result is obtained in Roundabout category (C12), which clearly illustrates that Roundabout category (C12) is the relatively easier category for the few-shot segmentation task. Meanwhile, our proposed algorithm obtains relatively bad results in helicopter class (C10), which indicates that it might be hard for our model to accurately match the query features with support prototypes in helicopter class (C10). By analyzing the helicopter (C10) in iSAID-5^i, we could find that the single size pattern and limited sample numbers of this class could contribute to this failure.

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TABLE VI
PERFORMANCE EVALUATION OF TINY OBJECTS

Methods	Split1	Split2	Split3	Mean
Panet [30]	1.4	1.5	2.1	1.7
Canet [27]	3.2	2.4	6.1	3.9
PMMs [32]	2.8	4.8	5.8	4.5
PFEnet [31]	2.2	4.3	6.3	4.3
Ours	**7.2**	**6.1**	**8.1**	**7.1**

The qualitative results are shown in Fig. 7, and the satisfied segmentation results demonstrate the strong detailed matching and scale-aware ability of our network. Particularly, it is clear that our model successfully parses the semantic objects in roundabout, ship, harbor, and large vehicle classes. Compared with the segmentation results from the PMMs, we could clearly find that our model is able to capture more semantic components and eliminate the background area. This phenomenon could be further explained by the reason that the detailed matching between the query features and support prototypes boosts the pixel-level identification ability and the scale-ware focal loss helps the model perceive more complete vision field capturing more details. However, compared with the ground truth, our parsing model does not acquire the pleasant segmentation results in the plane and the baseball diamond categories. This phenomenon could be explained by the reason that the plane class has relatively more complicate structures and diverse appearances, and the baseball diamond class generally would appear with more complicate background and many isolated semantic components.

Moreover, to further demonstrate the scale-aware ability of our proposed segmentation model, we perform several experiments to testify the segmentation performance on the tiny targets of the proposed networks. The overall experiments are performed with Resnet50 backbone in the one-shot setting and the experimental results are shown in Table VI. Compared with other algorithms, our model clearly has superior performance on tiny object. Particularly, our model acquires 7.1 mean mIOU over three splits. Aiming at directly analyzing the scale-aware ability of our proposed segmentation model, the qualitative results are shown in Fig. 8. These tiny and inconspicuous semantic objects generally would be surrounded by the complicate background or unimpressively appear in the corner. For instance, the large vehicle in the first column is surrounded by multiple unrelated foreground regions, which make the parsing model easily neglect the minute semantic objects. Meanwhile, the semantic objects in the following columns tend to appear in the corner of the scenario or locate in a place without salient shapes. In this manner, the parsing model normally lacks the ability to parse the semantic objects. However, with the help of our designed scale-aware focal loss, our model becomes aware of the sematic objects' scales. As shown in Fig. 8, the CAnet could hardly segment the tiny objects, especially for the large vehicle, where the CAnet could not parse anything. Meanwhile, the PAnet incorrectly recognizes some background as foreground and misses some obvious foreground areas. Compared with these methods, our model could effectively parse most foreground area of tiny objects with very little background area activated. This result clearly demonstrates the scale-aware ability of our proposed network.

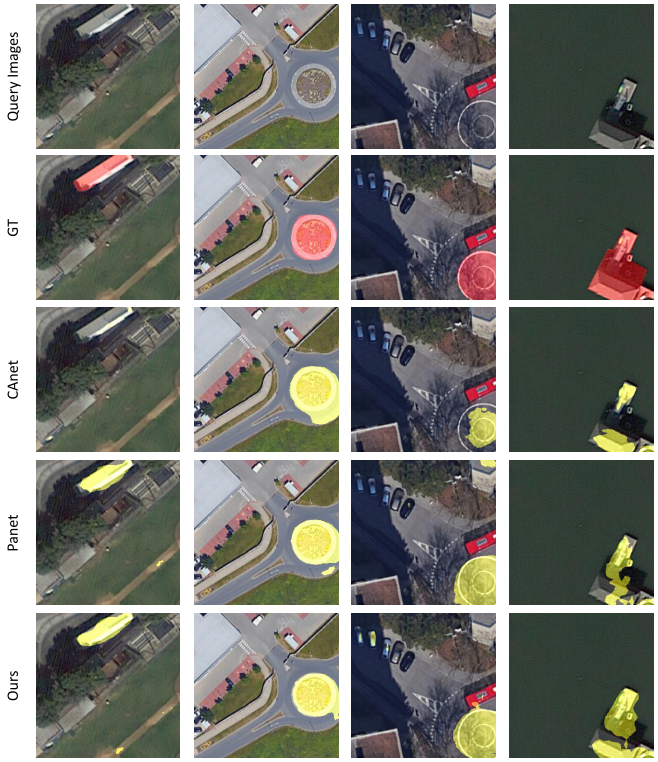


Fig. 8. Qualitative results of our model on the tiny and inconspicuous semantic objects.

TABLE VII
EFFICIENCY EVALUATION OF THE PROPOSED NETWORK

Methods	Params	FPS	
		1-shot	5-shot
VGG-16			
Panet [30]	18.3M	13.4	4.2
Canet [27]	26.4M	16.8	5.6
PMMs [32]	25.8M	18.7	9.4
PFEnet [31]	25.1M	40.3	36.9
Ours	25.8M	30.4	11.5
Resnet50			
Panet [30]	12.1M	13.6	4.6
Canet [27]	20.2M	16.2	5.5
PMMs [32]	19.6M	16.4	9.8
PFEnet [31]	34.4M	20.9	19.8
Ours	19.6M	29.2	11.9

Finally, to testify the efficiency of our proposed network, we collect the number of parameters and the frames per second (FPS) of diverse networks. The results are shown in Table VII. As shown in Table VII, the PFEnet has the highest FPS since it only uses the single prototype. Compared with the PFEnet, although our model has lower FPS because of the utilization of multiple prototypes, our model has relatively lower parameters. Moreover, the sacrifice of time provides the model with precise parsing guidance, which helps the model obtain better segmentation performance. In conclusion, our

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TABLE VIII

ABLATION STUDY ON THE PARAMETERS OF THE SCALE-AWARE LOSS (THE FIRST ROW DENOTES λ AND THE FIRST COLUMN DENOTES p)

Parameters p / λ	0.5	0.1	0.05	0.01
0	33.88	32.67	29.57	29.38
1	33.93	**34.29**	32.43	33.35
2	29.6	32.80	33.45	32.11
3	23.92	28.39	29.69	29.43

TABLE IX

ABLATION STUDY ON THE NUMBER OF GENERATED PROTOTYPES

2	3	4	5	Numbers
24.63	**28.19**	27.58	26.69	mIOU

TABLE X

ABLATION STUDY ON THE DESIGNED COMPONENTS OF THE PROPOSED ALGORITHM

Detailed Matching	Scale-aware Focal Loss	mIOU
		20.87
√		28.19
	√	28.46
√	√	**34.29**

model is able to tackle the few-shot segmentation task with satisfied efficiency.

D. Ablation Study

In order to further demonstrate the efficiency of our designed modules and further study the impact of the parameters in the scale-aware focal loss, we construct some ablation experiments in split 0 of iSAID-5^i with the one-shot setting and Resnet50 backbone. First, we set experiments to study the impact of p and λ. p is leveraged to exponentially bounce the impact of semantic objects' scale, and λ is directly added to modulate the effect of the semantic objects' scale. Thus, p must be a value that is above 0, and λ should be a value which is near 0. As illustrated in Table VIII, where the best performance is highlighted, the parsing performance would get the peak when λ equals 0.1 and p equals 1. Based on this result, we, respectively, set λ as 0.1 and p as 1 for the remaining experiments.

To rationally set the appropriate number of generated prototypes in the detailed match module, some experiments are elaborately performed with the single detailed matching. The results are shown in Table IX. Our model would get the best performance when the number of prototypes is set as 3. Naturally, we set the number of prototypes as 3 for the following experiments.

Additionally, we set a series of experiments to further, respectively, testify the designed modules of our proposed method, and the results are illustrated in Table X. It is clear that with the help of our designed detailed matching module, the parsing model obtains 7.32% mIOU improvement by accurately matching the query features and support prototypes at pixel level. Simultaneously, the scale-aware focal loss could provide the parsing model with 7.59% mIOU improvement via acquiring the adaptability to the scales of sematic objects. By effectively combining the two designed modules, the segmentation network obtains precise parsing guidance and becomes aware of the semantic objects' scale. Benefiting from these, the segmentation module could get 13.42% improvement in terms of mIOU.

V. CONCLUSION

In this article, we propose a novel scale-aware few-shot semantic segmentation network with detailed matching between the query and multiple support prototypes. Particularly, the detailed matching module would precisely match the query features and support prototypes with more accurate parsing guidance. Furthermore, the scale-aware focal loss would reinforce the scale-aware ability of the model to precisely parse the tiny and inconspicuous sematic objects. Extensive experiments are comprehensively conducted on the newly provided iSAID-5^i dataset, and our model achieves state-of-the-art performances in the elaborately designed datasets.

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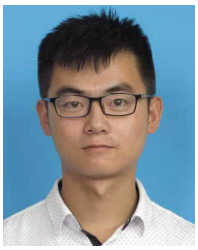
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